Relativistic Momentum

PHYS 2203 — Introductory Physics III

October 5, 2026

Overview

We finish Lecture 16's argument that classical momentum $m\mathbf{u}$ is not conserved in every frame. Requiring $\mathbf{F} = m\mathbf{a}$ in the particle's instantaneous rest frame then gives the relativistic force law and the momentum $\mathbf{p} = \gamma m\mathbf{u}$. Conserving $\mathbf{p}$ in every frame turns out to require a second conserved quantity, which Lecture 18 identifies as the energy.

1 Momentum Conservation (continued from Lecture 16)

The most useful law we have is conservation of momentum, $\mathbf{P}_{\text{total}} = \sum_i m_i \mathbf{u}_i$ = constant for an isolated system. The first postulate demands that such a law hold in *every* inertial frame, not just in one. That is a real constraint, and it is worth checking before assuming the law carries over.

Classically the check passes. Using the Galilean rule $u_i' = u_i - v$ of Lecture 13,

$$P'_{\text{total}} = \sum_i m_i(u_i - v) = \sum_i m_i u_i - v\sum_i m_i \qquad \Longrightarrow \qquad P'_{\text{total}} = P_{\text{total}} - v\sum_i m_i. \tag{1}$$

The two totals differ by a *constant*, since the system's mass does not change and v is fixed. So if P_{total} is unchanged by a collision, so is P'_{total}. That is exactly what the first postulate demands of a law of physics.

But the Galilean rule is wrong. Redo it with the relativistic velocity-addition rule of Lecture 16. Two particles of mass 4 and 3 (arbitrary units) collide head-on in frame O, as drawn. Their classical momenta are equal and opposite before *and* after, so $\sum_i m_i u_i = 0$ throughout and classical momentum is conserved in this frame.

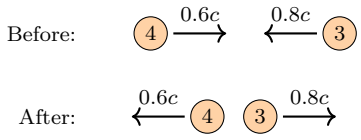


Figure 1: A collision constructed so that the *classical* total momentum vanishes in frame O, both before and after.

We take the collision to be elastic. Since O is the centre-of-momentum frame, each particle simply reverses its velocity.

Transform all four velocities into a frame O' moving at $v = 0.6c$, using $u' = \dfrac{u - v}{1 - vu/c^2}$:

$$\text{before, mass 4:} \quad u' = \frac{0.6c - 0.6c}{1 - (0.6)(0.6)} = 0, \tag{2}$$

$$\text{before, mass 3:} \quad u' = \frac{-0.8c - 0.6c}{1 - (-0.8)(0.6)} = \frac{-1.4c}{1.48} = -0.946c, \tag{3}$$

$$\text{after, mass 4:} \quad u' = \frac{-0.6c - 0.6c}{1 - (-0.6)(0.6)} = \frac{-1.2c}{1.36} = -0.882c, \tag{4}$$

$$\text{after, mass 3:} \quad u' = \frac{0.8c - 0.6c}{1 - (0.8)(0.6)} = \frac{0.2c}{0.52} = 0.385c. \tag{5}$$

Now add up the classical momenta in the new frame:

$$\text{before:} \quad \sum_i m_i u_i' = 4(0) + 3(-0.946c) = -2.84c, \tag{6}$$

$$\text{after:} \quad \sum_i m_i u_i' = 4(-0.882c) + 3(0.385c) = -2.38c. \tag{7}$$

The problem. $\sum_i m_i u_i$ is conserved in frame O but *not* in frame O'. A conservation law that holds in one inertial frame and fails in another violates the first postulate, so classical momentum $m\mathbf{u}$ cannot be correct.

So $\mathbf{p} = m\mathbf{u}$ must be replaced. Notice what the argument does *not* tell us: it says the classical expression is wrong, but not what the right one is. We will derive it now.

Experiments show that $\mathbf{F} = m\mathbf{a}$ holds when $u' \ll c$, including when $u' = 0$ instantaneously (the O' frame), where u' is the velocity of the object in O'. *Does an observer in the O frame agree that $\mathbf{F} = m\mathbf{a}$ is valid?*

2 Transforming the Acceleration

Let u be the velocity of the object in O and u' its velocity in O', with v the constant relative speed of the two frames, as in Lecture 16. We want $a' = \mathrm{d}u'/\mathrm{d}t'$, the acceleration measured in O'.

Step 0: use the chain rule. The acceleration in O' is $a' = \mathrm{d}u'/\mathrm{d}t'$. By the chain rule,

$$a' = \frac{\mathrm{d}u'}{\mathrm{d}t'} = \frac{\mathrm{d}u'}{\mathrm{d}t}\,\frac{\mathrm{d}t}{\mathrm{d}t'} = \frac{\mathrm{d}u'/\mathrm{d}t}{\mathrm{d}t'/\mathrm{d}t}. \tag{8}$$

Both pieces are now derivatives with respect to t, which is what our formulas are written in terms of.

Step 1: the numerator, $\mathrm{d}u'/\mathrm{d}t$. The quantity u' depends on t only through u, so by the chain rule

$$\frac{\mathrm{d}u'}{\mathrm{d}t} = \frac{\mathrm{d}u'}{\mathrm{d}u}\,\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}u'}{\mathrm{d}u}\,a, \tag{9}$$

where $a = \mathrm{d}u/\mathrm{d}t$ is the acceleration in O. So everything reduces to finding $\mathrm{d}u'/\mathrm{d}u$.

Step 2: differentiate the velocity-addition formula. From Lecture 16,

$$u' = \frac{u - v}{1 - uv/c^2}. \tag{10}$$

This is a quotient, so we use the **quotient rule**: if $f = g/h$ then

$$\frac{\mathrm{d}f}{\mathrm{d}u} = \frac{g'h - gh'}{h^2}, \tag{11}$$

where $'$ denotes $\mathrm{d}/\mathrm{d}u$. Identify the two pieces, remembering that v and c are *constants* and only u varies:

$$g = u - v, \qquad g' = 1, \qquad\qquad h = 1 - \frac{uv}{c^2}, \qquad h' = -\frac{v}{c^2}. \tag{12}$$

Substituting,

$$\frac{\mathrm{d}u'}{\mathrm{d}u} = \frac{(1)\left(1 - \dfrac{uv}{c^2}\right) - (u - v)\left(-\dfrac{v}{c^2}\right)}{\left(1 - \dfrac{uv}{c^2}\right)^2}. \tag{13}$$

Work on the numerator alone. The two minus signs in the second term make it positive:

$$\begin{aligned} \text{numerator} &= 1 - \frac{uv}{c^2} + \frac{v(u - v)}{c^2} \\ &= 1 - \frac{uv}{c^2} + \frac{uv}{c^2} - \frac{v^2}{c^2} \\ &= 1 - \frac{v^2}{c^2}. \end{aligned} \tag{14}$$

The two uv/c^2 terms cancel exactly, and what survives is $1 - v^2/c^2 = 1/\gamma_v^2$. Hence

$$\boxed{\frac{\mathrm{d}u'}{\mathrm{d}u} = \frac{1}{\gamma_v^2\,(1 - uv/c^2)^2}} \tag{15}$$

Step 3: the denominator, $\mathrm{d}t'/\mathrm{d}t$. From the Lorentz transformation, $t' = \gamma_v\left(t - \dfrac{v}{c^2}x\right)$. Differentiate with respect to t, treating γ_v and v as constants and using $\mathrm{d}x/\mathrm{d}t = u$:

$$\frac{\mathrm{d}t'}{\mathrm{d}t} = \gamma_v\left(\frac{\mathrm{d}t}{\mathrm{d}t} - \frac{v}{c^2}\frac{\mathrm{d}x}{\mathrm{d}t}\right) = \gamma_v\left(1 - \frac{uv}{c^2}\right). \tag{16}$$

Step 4: put the pieces together. Divide Step 1 by Step 3:

$$a' = \frac{a}{\gamma_v^2\,(1 - uv/c^2)^2\,\gamma_v\,(1 - uv/c^2)} = \frac{a}{\gamma_v^2\,(1 - uv/c^2)^2} \cdot \frac{1}{\gamma_v\,(1 - uv/c^2)}, \tag{17}$$

and collecting the like factors gives the result:

$$\boxed{a' = \frac{a}{\gamma_v^3\,(1 - uv/c^2)^3}} \tag{18}$$

Notice how different this is from the Galilean case, where $a' = a$ and acceleration is the same in every frame. Here it is not, and the discrepancy depends on the object's own velocity u as well as on the frame speed v.

3 The Relativistic Force Law and Momentum

Let us choose $v = u$ (the object is at zero velocity instantaneously in the O' frame) for three reasons:

1. We can assume $\mathbf{F}' = m\mathbf{a}'$ (zero velocity).
2. We can assume $\mathbf{F}' = \mathbf{F}$ when the motion is in the x direction (will be derived in E&M 2).
3. It simplifies the equation for a'.

Setting $v = u$ in Eq. (18) makes it collapse. Write γ for the Lorentz factor built from the particle's *own* speed,

$$\gamma \equiv \frac{1}{\sqrt{1 - u^2/c^2}}, \qquad \text{so that} \qquad 1 - \frac{uv}{c^2} \;\longrightarrow\; 1 - \frac{u^2}{c^2} = \frac{1}{\gamma^2}. \tag{19}$$

Substituting that into Eq. (18), with $\gamma_v \to \gamma$ as well,

$$a' = \frac{a}{\gamma^3 \left(1/\gamma^2\right)^3} = \frac{a}{\gamma^3/\gamma^6} = \gamma^3\, a. \tag{20}$$

Only the particle's own speed u appears from here on — there is no second frame left to track.

Using $F' = ma'$ and $F = F'$,

$$\frac{F}{m} = \gamma^3 a. \tag{21}$$

We would like to write this as $F = \dfrac{\mathrm{d}}{\mathrm{d}t}(\text{something})$, to draw an analogy with $F = \mathrm{d}p/\mathrm{d}t$ from classical mechanics. It turns out that the right-hand side is a total time derivative:

$$\gamma^3 a = \frac{\mathrm{d}}{\mathrm{d}t}\left(\gamma\, u\right). \tag{22}$$

You will verify this on Homework 6 with the product rule, and the chain rule applied to $\gamma = (1 - u^2/c^2)^{-1/2}$. As a quick check, at low speed $\gamma \approx 1$ and both sides reduce to a.

Relativistic force. Therefore

$$\boxed{F = \frac{\mathrm{d}}{\mathrm{d}t}\left(\gamma\, mu\right)} \qquad [\neq ma\,!] \tag{23}$$

This looks like the classical relation $\mathbf{F} = \dfrac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$ (force equals rate of change of momentum), but with the **relativistic momentum**

$$\boxed{\mathbf{p} = \gamma\, m\mathbf{u}.} \tag{24}$$

- $\mathbf{p} \approx m\mathbf{u}$ when $u \ll c$ (recovers the classical relation).
- $p \to \infty$ as $u \to c$.

Conservation of relativistic momentum. Experiment shows that in every collision and reaction ever measured, the total relativistic momentum is unchanged. For a system of N particles,

$$\sum_{i=1}^{N} \mathbf{p}_i \quad \text{is unchanged by collisions and reactions between particles.} \tag{25}$$

Does this repair the frame problem? Section 1 showed that classical momentum, conserved in one frame, is not conserved in another. We now ask the same question of $\mathbf{p} = \gamma m \mathbf{u}$: if $\sum_i \gamma m_i u_i$ is conserved in frame O, is it also conserved in frame O'?

Notation. Three speeds now appear, and each has its own Lorentz factor:

* v, the speed of O' relative to O, with $\gamma_v = 1/\sqrt{1 - v^2/c^2}$;
* u, a particle's velocity measured in O, with $\gamma_u = 1/\sqrt{1 - u^2/c^2}$;
* u', the same particle's velocity measured in O', with $\gamma_{u'} = 1/\sqrt{1 - u'^2/c^2}$.

The momentum each observer assigns uses *that observer's* measured velocity: $p = \gamma_u m u$ in O, and $p' = \gamma_{u'} m u'$ in O'. To compare the two we must write $\gamma_{u'}$ and u' in terms of u and v.

Step 1: find $\gamma_{u'}$. Substituting the velocity-addition formula $u' = (u - v)/(1 - uv/c^2)$ into $\gamma_{u'} = 1/\sqrt{1 - u'^2/c^2}$ and simplifying gives

$$\boxed{\gamma_{u'} = \gamma_v \, \gamma_u \left(1 - \frac{uv}{c^2}\right).} \tag{26}$$

You will derive this on Homework 6.

Step 2: find $\gamma_{u'} u'$. Multiply the result of Step 1 by the velocity-addition formula. The factor $\left(1 - uv/c^2\right)$ appears on top in one and on the bottom in the other, so it cancels:

$$\gamma_{u'} \, u' = \gamma_v \gamma_u \left(1 - \frac{uv}{c^2}\right) \cdot \frac{u - v}{1 - uv/c^2} = \gamma_v \left(\gamma_u u - v \, \gamma_u\right). \tag{27}$$

Step 3: sum over particles. Multiply by m_i and add up all the particles. The first term gives $\gamma_v \sum_i \gamma_{u_i} m_i u_i = \gamma_v P_{\text{total}}$, so

$$\boxed{P'_{\text{total}} = \gamma_v P_{\text{total}} - v \, \gamma_v \sum_i \gamma_{u_i} m_i.} \tag{28}$$

Compare this with the classical version in Section 1. There the second term was $v \sum_i m_i$, a genuine constant. Here it is $v\gamma_v \sum_i \gamma_{u_i} m_i$, and $\sum_i \gamma_{u_i} m_i$ is not obviously constant — each γ_{u_i} depends on a particle speed that changes during the collision. So P'_{total} is conserved *if and only if* $\sum_i \gamma_{u_i} m_i$ is conserved as well.

Momentum conservation needs a partner. For relativistic momentum to be conserved in *every* inertial frame, the quantity $\sum_i \gamma_{u_i} m_i$ must also be conserved. The first postulate therefore predicts a *second* conservation law before we have even looked for one. The next lecture identifies the quantity: multiplied by c^2, it is the total energy.

We assumed that m is independent of inertial reference frame. This is an experimental fact. (The charge q is also independent of frame.)